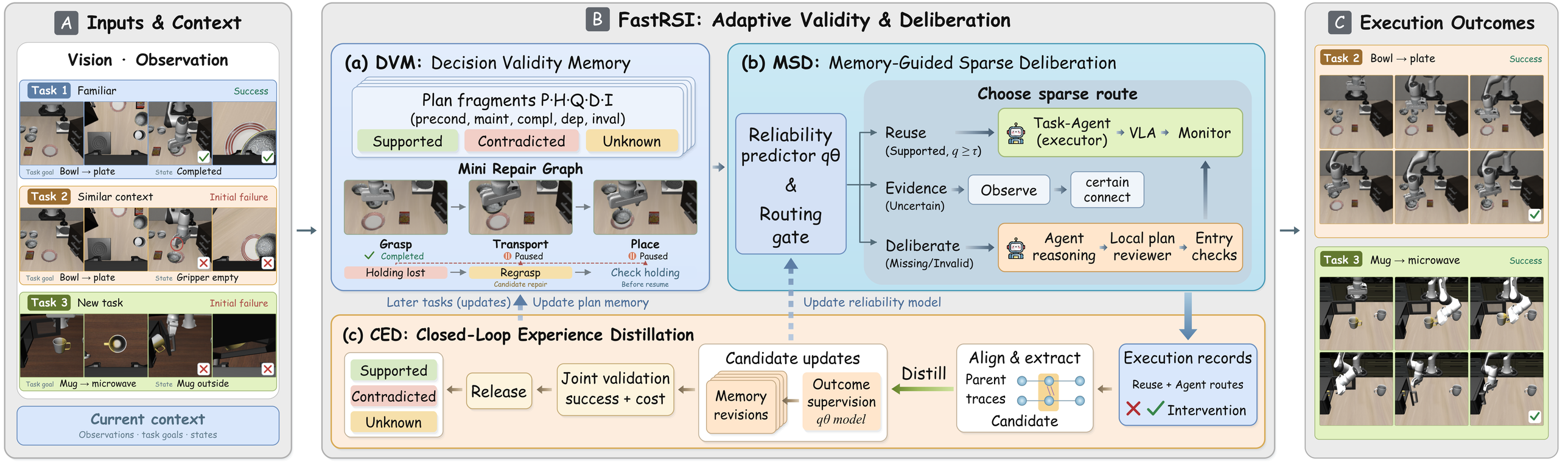

A Inputs & Context
Vision · Observation
Task 1 Familiar
Success
Task goal Bowl → plate
State Completed
Task 2 Similar context
Initial failure
Task goal Bowl → plate
State Gripper empty
Task 3 New task
Initial failure
Task goal Mug → microwave
State Mug outside
Current context
Observations · task goals · states
B FastRSI: Adaptive Validity & Deliberation
(a) DVM: Decision Validity Memory
Plan fragments P·H·Q·D·I
(precond, maint, compl, dep, inval)
Supported
Contradicted
Unknown
Mini Repair Graph
Grasp
Completed
Transport
Paused
Place
Paused
Holding lost
Regrasp
Candidate repair
Check holding
Before resume
(b) MSD: Memory-Guided Sparse Deliberation
Choose sparse route
Reliability predictor qθ & Routing gate
Reuse
(Supported, q ≥ τ)
Task-Agent (executor)
VLA
Monitor
Evidence
(Uncertain)
Observe
certain connect
Deliberate
(Missing/Invalid)
Agent reasoning
Local plan reviewer
Entry checks
Later tasks (updates)
Update plan memory
Update reliability model
(c) CED: Closed-Loop Experience Distillation
Supported
Contradicted
Unknown
Release
Joint validation success + cost
Candidate updates
Memory revisions
Outcome supervision
qθ model
Distill
Align & extract
Parent traces
Candidate
Execution records
Reuse + Agent routes
Intervention
C Execution Outcomes
Task 2 Bowl → plate
Success
Task 3 Mug → microwave
Success